

HSC Investigating Science - The Definitive Exam-Mapped Notes (v2)

A Comprehensive Guide Directly Linked to the Full Syllabus and Past HSC Questions

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Introduction: How to Use These Notes

This guide is your definitive, data-driven resource for the HSC Investigating Science exam. It has been generated directly from a comprehensive Knowledge Graph, ensuring every syllabus dot point is covered and linked to real-world examples from past HSC papers where applicable.

Study Strategy: 1. **Systematic Coverage:** The notes are structured sequentially from Module 1 to Module 8. Work through them systematically to build a strong foundational knowledge. 2. **Focus on Inquiry Questions:** Each module is framed by inquiry questions. Use these to guide your understanding and test your knowledge. 3. **Connect Concepts to Assessments:** For Year 12 modules, pay close attention to the “HSC Example” provided. This shows you exactly how NESA assesses that concept. 4. **Master the Glossary:** The definitions at the end are your key to unlocking marks. They are precise and written for exam conditions.

Year 11 Foundational Content

Module 1: Cause and Effect – Observing

Inquiry question: How does observation instigate scientific investigation?

- **concept:M1-1-1:** Carry out a practical investigation to record both quantitative and qualitative data from observations.
- **concept:M1-1-2:** Discuss and evaluate the characteristics of observations made compared to inferences drawn in respect of the practical investigation.
- **concept:M1-1-3:** Research how observation has instigated experimentation to investigate cause and effect in historical examples.
- **concept:M1-1-4:** Assess ways in which Aboriginal and Torres Strait Islander Peoples use observation to develop an understanding of Country and Place in order to create innovative ways of managing the natural environment.

Inquiry question: What are the benefits and drawbacks of qualitative and quantitative observations?

- **concept:M1-2-1:** Carry out a practical activity to qualitatively and quantitatively describe.
- **concept:M1-2-2:** Analyse the quantitative data from information sources.

- **concept:M1-2-3:** Evaluate the differences between qualitative and quantitative observations and data and where these are used.

Inquiry question: How does primary data provide evidence for further investigation?

- **concept:M1-3-1:** Use data gathered to plan a practical investigation.
- **concept:M1-3-2:** Develop a method to collect primary data for a practical investigation.
- **concept:M1-3-3:** Evaluate how observation is limited by the observational tools available.

Inquiry question: How does the collection and presentation of primary data affect the outcome of a scientific investigation?

- **concept:M1-4-1:** Carry out the planned practical investigation to collect primary data.
- **concept:M1-4-2:** Apply conventions for collecting and recording observations to qualitatively and quantitatively analyse the primary data.
- **concept:M1-4-3:** Compare the usefulness of observations recorded in the initial practical activity with the primary data gathered in this planned practical investigation.

Inquiry question: How do conclusions drawn from the interpretation of primary data promote further scientific investigation?

- **concept:M1-5-1:** Draw conclusions from the analysis of the primary data collected in the practical investigation.
- **concept:M1-5-2:** Evaluate the process of drawing conclusions from the primary data collected.
- **concept:M1-5-3:** Assess the findings of the scientific investigation in relation to the findings of other related investigations.
- **concept:M1-5-4:** Assess the need to make further observations by gathering data about other phenomena arising from the practical investigation.

Module 2: Cause and Effect – Inferences and Generalisations

Inquiry question: What inferences can be drawn from observations?

- **concept:M2-1-1:** Investigate the practices of Aboriginal and Torres Strait Islander Peoples that relate to observations and inferences.

- **concept:M2-1-2:** Conduct a collaborative practical investigation and collect a range of qualitative and quantitative primary data.
- **concept:M2-1-3:** Make inferences and conclusions derived from the primary data collected in this collaborative practical investigation.

Inquiry question: How is secondary-sourced data used in practical investigations?

- **concept:M2-2-1:** Collect qualitative and quantitative secondary-sourced data to validate the inferences and conclusions drawn from the practical investigation.
- **concept:M2-2-2:** Discuss how secondary-sourced data adds to the inferences and conclusions drawn from primary data.
- **concept:M2-2-3:** Evaluate the usefulness of considering secondary-sourced research before undertaking an investigation to collect primary data.

Inquiry question: How does humans' ability to recognise patterns affect the way they interpret data?

- **concept:M2-3-1:** Describe patterns that have been observed over time throughout the Universe and in nature.
- **concept:M2-3-2:** Interpret data in order to propose a hypothesis based on an irregular pattern observed over time.
- **concept:M2-3-3:** Examine the human tendency to observe patterns and misinterpret information.
- **concept:M2-3-4:** Discuss how the tendency to recognise patterns, even when they may not exist, can lead to misinterpretation of data.
- **concept:M2-3-5:** Discuss the role and significance of outliers in data.

Inquiry question: How can hypotheses and assumptions be tested?

- **concept:M2-4-1:** Gather secondary-sourced data describing historical instances of long-standing assumptions that have been updated by scientific investigation.
- **concept:M2-4-2:** Propose an inquiry question, construct a hypothesis and conduct an investigation that tests a common assumption.
- **concept:M2-4-3:** Use appropriate representations to analyse the data.

Inquiry question: What generalisations and assumptions are made from observed data?

- **concept:M2-5-1:** Make generalisations to describe any trends found in the data.
- **concept:M2-5-2:** Draw conclusion based on generalisations.

Inquiry question: What role do peers play in scientific investigation?

- **concept:M2-6-1:** Assess the input that collaborative teams and alternative perspectives have had on the development of hypotheses and research questions.
 - **concept:M2-6-2:** Assess the scientific community's current understanding of scientific mysteries and outline why this understanding remains incomplete.
 - **concept:M2-6-3:** Evaluate biases that may have affected the scientific thinking of European settlers about Aboriginal and Torres Strait Islander Peoples' ecological understanding and knowledge.
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Module 3: Scientific Models

Inquiry question: What is a scientific model?

- **concept:M3-1-1:** Examine the types of models that may be used in science (diagrams, physical replicas, mathematical representations, analogies, computer simulations).

Inquiry question: What makes scientific models useful?

- **concept:M3-2-1:** Examine the use of scientific models (e.g., epidemic, Universe, atomic, climate models).
- **concept:M3-2-2:** Outline how models have been used to illustrate, simplify and represent scientific concepts and processes.
- **concept:M3-2-3:** Explain how scientific models are used to make predictions that are difficult to analyse in the real world.
- **concept:M3-2-4:** Assess the effectiveness of models at facilitating the understanding of scientific processes, structures and mathematical relationships.
- **concept:M3-2-5:** Evaluate how scientific models draw on a growing body of data to refine predictions and test new hypotheses.

Inquiry question: When should a particular model be used?

- **concept:M3-3-1:** Explain why new evidence can challenge the use of existing scientific models, leading to them being contested and refined or replaced.
- **concept:M3-3-2:** Compare the limitations of simple and complex scientific models.

Inquiry question: How can a model be constructed to simplify understanding of a scientific concept?

- **concept:M3-4-1:** Investigate a scientific concept or process that can be represented using a model.
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Module 4: Theories and Laws

Inquiry question: What are the differences and similarities between scientific theories and laws?

- **concept:M4-1-1:** Collect primary data to investigate the law of conservation of mass.
- **concept:M4-1-2:** Collect secondary-sourced data to investigate the theory of plate tectonics.
- **concept:M4-1-3:** Compare the characteristics of theories and laws.

Inquiry question: What leads to a theory being developed?

- **concept:M4-2-1:** Gather secondary-sourced data to investigate the supporting evidence and development of theories (e.g., germ theory, oxygen theory of combustion).
- **concept:M4-2-2:** Gather secondary-sourced data to investigate how aspects of a theory can be disproved (e.g., Geocentric Theory, Lamarck's Theory of Inheritance).

Inquiry question: What leads to the acceptance of a scientific law?

- **concept:M4-3-1:** Gather secondary-sourced data to investigate and assess the evidence that supports scientific laws (e.g., Newton's Second Law, Avogadro's Law).
- **concept:M4-3-2:** Design and collect primary data to show that results can be predicted by laws (e.g., Ohm's Law, Law of Conservation of Energy).

Inquiry question: How are theories and laws used in science?

- **concept:M4-4-1:** Investigate how the law of conservation of energy is applied in different science disciplines.
- **concept:M4-4-2:** Demonstrate, using evidence and examples, how diverse phenomena have been unified into specific theories (e.g., atomic theory, theory of evolution).
- **concept:M4-4-3:** Investigate how investigations of nuclear reactions changed the interpretation of the laws of conservation of mass and energy.

Year 12 High-Priority Content

Module 5: Scientific Investigations

Inquiry question: What initiates an investigation?

- **concept:M5-1-1:** Research the factors that led scientists to investigate phenomena (e.g., Marshall and Warren - peptic ulcers; Von Helmont - plant growth; Spencer - microwaves).
- **concept:M5-1-2:** Propose a reason for scientists undertaking their investigations by examining the type of data they sought.
- **concept:M5-1-3:** Determine the hypotheses that were tested in each scientific investigation.
- **concept:M5-1-4:** Describe where deviations from the traditional scientific method were necessary.

Inquiry question: What type of methodology best suits a scientific investigation?

- **concept:M5-2-1:** Evaluate the objectives and data collected in an investigation by a recognised scientist (e.g., Eratosthenes and Earth's circumference).
- **concept:M5-2-2:** Evaluate the methodology of scientific investigations.
- **concept:M5-2-3:** Develop a method most appropriate to test a hypothesis following observation.
- **concept:M5-2-4:** Justify the type of methodology used to test the hypothesis.

Inquiry question: How is the integrity of a scientific investigation judged?

- **concept:M5-3-1:** Evaluate the design of a student investigation.
- **concept:M5-3-2:** Conduct the planned investigation and collect, record and analyse primary data.
- **concept:M5-3-3:** Draw a conclusion and suggest further investigation.
- **concept:M5-3-4: Evaluate the validity of the investigation.**
 - **Definition:** An experiment is valid if its method is appropriate and directly tests the intended hypothesis. Core components include controlling variables, having a single independent variable, and appropriate measurement of the dependent variable.

- **HSC Example (2021 Q22b - GMO Mice):** An experiment tested if genetically modified mice on a high-fat diet gained more weight. To be valid, it required four groups to isolate the effects of the gene and the diet, including a control group (normal mice, normal diet) and a hypothesis test group (GMO mice, high-fat diet).
- **concept:M5-3-5: Evaluate the reliability of the investigation.**
 - **Definition:** An experiment is reliable if the results are consistent and repeatable. This is achieved through repetition (at least 3 trials) and using a sufficiently large sample size.
 - **HSC Example (2020 Q25a - Self-Injection):** A researcher injected herself with a bacterium to see if it caused a disease. This experiment is unreliable because the sample size is only one, making it impossible to know if the result is consistent or a one-off personal reaction.

Inquiry question: What is the structure of an investigative report?

- **concept:M5-4-1: Review a published and peer-reviewed scientific report to determine the conventions of writing.**
 - **Definition:** A formal scientific report follows a clear structure (Aim, Hypothesis, Method, Results, Discussion, Conclusion). The language must be objective, in the third person, and use precise scientific terminology.
 - **HSC Example (2023 Q30 - Student Report):** A student's report was criticised for using informal headings like 'My prediction is' instead of 'Hypothesis' and for describing results with colloquial language ('spurt height') instead of quantitative data in a table.
- **concept:M5-4-2:** Use a sample of a published secondary source to identify its purpose, language style, and structure.
- **concept:M5-4-3:** Compare and contrast the structures and functions of a scientific investigation and its written report.
- **concept:M5-4-4:** Prepare a report on the student investigation that was carried out.

Module 6: Technologies

Inquiry question: How does technology enhance and/or limit scientific investigation?

- **concept:M6-1-1:** Design a practical investigation that uses available technologies to measure variables.

- **concept:M6-1-2:** Conduct the practical investigation and evaluate the limitations of the technologies used.
- **concept:M6-1-3:** Investigate measuring devices and assess the likelihood of random and systematic errors and their accuracy.
- **concept:M6-1-4:** Compare the accuracy of analogue and digital technologies.
- **concept:M6-1-5:** Assess the safety of technologies selected for an investigation.

Inquiry question: How have developments in technology led to advances in scientific theories and laws?

- **concept:M6-2-1: Assess the impact that developments in technologies have had on the accumulation of evidence for scientific theories, laws and models.**
 - **Definition:** Science and technology have a feedback loop. New technology enables new scientific understanding, which in turn enables the development of even newer technology.
 - **HSC Example (2024 Q36 - DNA & GMOs):** X-ray crystallography (technology) was essential for Franklin to produce Photo 51. This image (science) allowed Watson and Crick to develop the double helix model of DNA. This new understanding enabled the development of recombinant DNA technology to create GMOs (new technology).
- **concept:M6-2-2:** Assess the impact that developments in scientific theories have had on new technologies.
- **concept:M6-2-3:** Investigate the value of Aboriginal and Torres Strait Islander Peoples' knowledge of medicinal and material uses of plants.

Module 7: Fact or Fallacy?

Inquiry question: How can a claim be tested?

- **concept:M7-1-1:** Plan and conduct an investigation based on testing a claim.
- **concept:M7-1-2:** Evaluate the impact of sample selection and sample sizes on results.
- **concept:M7-1-3:** Compare emotive advertising with evidence-based claims.

Inquiry question: What factors can affect the way data can be interpreted, analysed and understood?

- **concept:M7-2-1:** Justify the use of placebos, double-blind trials and control groups.
- **concept:M7-2-2:** Evaluate the impact of societal and economic influences on the collection and interpretation of data.

Inquiry question: What type of evidence is needed to draw valid conclusions?

- **concept:M7-3-1: Evaluate how evidence of a correlation can be misinterpreted as causation.**
 - **Definition:** Correlation is a relationship where two variables change together, but it does not imply one causes the other. Causation is where a change in one variable directly causes a change in another. Mistaking correlation for causation is a common logical fallacy.
 - **HSC Example (2019 Q25c - Whales & CO):** A graph showed that as whale sightings increased, atmospheric CO₂ also increased. A student incorrectly infers that the whales cause the increase in CO₂. This is a correlation, not causation, as a third factor (e.g., global economic growth) could be responsible for both trends.

Inquiry question: How does the reporting of science influence the general public's understanding?

- **concept:M7-4-1:** Examine a contemporary scientific debate and how it is portrayed in the mainstream media.
- **concept:M7-4-2:** Evaluate the use and interpretation of the terms ‘theory’, ‘hypothesis’, ‘belief’ and ‘law’ in media reporting.
- **concept:M7-4-3:** Compare the difference in reporting between a peer-reviewed journal article and a popular media article.
- **concept:M7-4-4:** Analyse how conflicts of interest can result in scientific evidence being suppressed or misrepresented.
- **concept:M7-4-5:** Describe the halo effect and how it can result in the rejection of valid alternative perspectives.
- **concept:M7-4-6: Analyse a pseudo-scientific claim and how scientific language can be manipulated.**
 - **Definition:** Pseudoscience is a claim, belief, or practice presented as scientific, but which does not adhere to the scientific method. It often uses scientific-sounding language to mislead.

- **HSC Example (2020 Q27a - Astrology):** Astrologists use scientific-sounding terms like ‘retrograde motion’ to give their claims an appearance of scientific legitimacy, misleading the public into believing astrology is based on scientific evidence when it is not testable or falsifiable.

Inquiry question: Can the scientific community and process of peer review find ‘the truth’?

- **concept:M7-5-1:** Research a scientist who has falsified their results and discuss the process used to uncover the fraud.
 - **concept:M7-5-2:** Analyse the scientific debate surrounding ‘publication’ and the implications of ‘publish or perish’.
 - **concept:M7-5-3:** Evaluate the feasibility of science to effectively manage, review, and validate the increasing volume of scientific papers.
 - **concept:M7-5-4: Analyse the benefits of peer review in relation to the advancement of science.**
 - **Definition:** Peer review is the evaluation of scientific work by other experts in the same field to ensure its quality and validity before it is published. It acts as a critical quality control filter.
 - **HSC Example (2019 Q29b - Marshall & Warren):** Marshall and Warren’s paper on *H. pylori* causing ulcers was peer-reviewed before publication. This process ensured their revolutionary idea was scrutinised for validity and methodological rigour, which helped it gain acceptance and advance medicine.
 - **concept:M7-5-5:** Discuss the impact of fake science journals on the public perception of science.
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Module 8: Science and Society

Inquiry question: How do science-related events affect society’s view of science?

- **concept:M8-1-1:** Investigate case studies of past events to consider how they have affected the public image of science (e.g., nuclear meltdowns, vaccine development).

Inquiry question: Why is scientific research regulated?

- **concept:M8-2-1:** Investigate the need for the regulation of scientific research (e.g., genetic modification, biotechnological weaponry).
- **concept:M8-2-2: Investigate and assess ethical issues surrounding current scientific research.**
 - **Definition:** Ethical research is governed by core principles (NHMRC): Informed Consent (participants are briefed on risks), Beneficence (maximise benefits, minimise harm), and Justice (fair selection of participants).
 - **HSC Example (2020 Q25b - Ethics Committee):** For a researcher who self-injected a bacterium, an ethics committee would likely require prior investigation on animal models to decrease the risk of human harm and would enforce strict animal welfare protocols.
- **concept:M8-2-3:** Investigate a range of international scientific codes of conduct.
- **concept:M8-2-4:** Evaluate the effectiveness of international regulation in scientific research.

Inquiry question: How do economic, social and political influences affect scientific research?

- **concept:M8-3-1:** Evaluate the costs involved in space exploration compared to investments in social issues.
- **concept:M8-3-2:** Evaluate how scientific research aids economic development and human progress.
- **concept:M8-3-3:** Evaluate the impacts of scientific research on world health and human wellbeing.
- **concept:M8-3-4: Analyse the impacts that governments and large corporations have on scientific research.**
 - **Definition:** Economic and political forces significantly influence science. A conflict of interest occurs when a researcher's private interests (e.g., financial ties to a company) could improperly influence their professional judgment.
 - **HSC Example (2022 Q33b - Corporate Funding & 2020 Q32c - Climate Change):** Increased corporate funding for science can create pressure for results favourable to the funder's products. An astrophysicist funded by oil companies claiming water vapour is the main cause of climate change is a major conflict of interest, as the fossil fuel industry has a financial stake in downplaying CO₂'s role.
- **concept:M8-3-5: Evaluate how personal, cultural and socioeconomic perspectives can influence the direction of scientific research.**
 - **Definition:** Social and political factors influence science. Scientific evidence can be used to inform political decisions and shape public opinion.

- **HSC Example (2019 Q30b - Franklin River Dam):** Scientists provided evidence on the negative environmental impacts of a proposed dam. This evidence was used by activists (social factor) and ultimately influenced a High Court decision to stop the dam (political outcome), demonstrating the interplay between science and society.

Complete Glossary of Key Terms

Term	Definition
Accuracy	How close a measured value is to the true or accepted value.
Bias	A systematic error or prejudice that can influence the results of an investigation, leading to a conclusion that is not objective.
Causation	The relationship where a change in one variable is directly responsible for a change in another variable.
Conflict of Interest	A situation where competing interests (e.g., financial) could potentially compromise a researcher's impartiality or objectivity in conducting or reporting research.
Control Group	A group in an experiment that is not exposed to the independent variable, used as a baseline to compare the results of the experimental group.
Controlled Variable	A variable that is kept constant throughout an experiment to ensure a fair test.
Correlation	A mutual relationship or connection between two or more things, where one variable changes in relation to another.
Dependent Variable	The variable that is measured or observed in response to the change in the independent variable.
Double-Blind Trial	An experiment in which neither the participants nor the researchers know who is receiving a particular treatment, used to prevent bias.

Term	Definition
Ethics	Moral principles that govern a person's or group's behaviour, concerning what is right and wrong in the conduct of scientific research.
Hypothesis	A testable prediction about the relationship between variables, which can be supported or refuted by evidence from an investigation.
Independent Variable	The variable that is deliberately changed or manipulated by the investigator.
Inference	A logical conclusion based on observations and prior knowledge.
Law (Scientific)	A statement that describes an observed phenomenon or a pattern in nature, often expressed as a mathematical relationship, without explaining why it occurs.
Model (Scientific)	A simplified representation of a system, object, or phenomenon that helps scientists understand and predict its behaviour.
Observation	The process of gathering information about events or processes in a careful, orderly way, using the senses or scientific instruments.
Outlier	A data point that is significantly different from other observations in a data set.
Peer Review	The evaluation of scientific work by other experts in the same field to ensure its quality and validity before it is published.
Placebo	A substance or treatment with no active therapeutic effect, used as a control in clinical trials to test the effectiveness of a new treatment.
Precision	How close multiple measurements of the same quantity are to each other.
Pseudoscience	A claim, belief, or practice presented as scientific, but which does not adhere to the scientific method and lacks supporting evidence.
Qualitative Data	Descriptive data that is observed, not measured (e.g., colour, texture, smell).
Quantitative Data	Numerical data that is measured or counted (e.g., length, mass, time, temperature).

Term	Definition
Random Error	Unpredictable variations in measurements that occur by chance and affect the precision of the results.
Reliability	The extent to which the results of an experiment are consistent and repeatable.
Systematic Error	An error that is consistent and repeatable, often caused by faulty equipment or a flawed experimental method, which affects the accuracy of the results.
Technology	The application of scientific knowledge for practical purposes, especially in industry, involving the development of devices, methods, and systems.
Theory (Scientific)	A well-substantiated explanation of some aspect of the natural world, based on a body of facts that have been repeatedly confirmed through observation and experiment.
Validity	The extent to which an experiment tests the intended hypothesis (i.e., it is a 'fair test').